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Applying the technology acceptance model to examine factors influencing chatbot use in HPV prevention education among nursing students

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Abstract

Background Human papillomavirus (HPV) is one of the most common sexually transmitted infections worldwide. However, its coverage in nursing curricula is often limited. Chatbots offer a novel approach to enhance digital health education, yet their acceptance among nursing students remains underexplored.

Objective This study examined nursing students' acceptance and use of an HPV prevention chatbot, based on the Technology Acceptance Model (TAM).

Methods A cross-sectional study was conducted with 210 nursing students in central Taiwan. Participants engaged with a LINE-based chatbot that included thematic modules, an HPV knowledge assessment, and an expert Q&A. Data were analyzed using descriptive statistics, *t*-tests, ANOVA, Pearson correlation, and linear regression. Open-ended feedback was also analyzed.

Results Perceived ease of use significantly predicted perceived usefulness ($\beta = 0.859$, $p < 0.001$). Both predicted attitude ($R^2 = 82.2\%$), which, along with usefulness, predicted behavioral intention ($R^2 = 90.8\%$). Behavioral intention strongly predicted actual use ($\beta = 0.933$, $p < 0.001$, $R^2 = 87.1\%$). Age, prior chatbot experience, and number of sexual partners significantly influenced perceived usefulness and perceived ease of use.

Conclusion The chatbot was well accepted and effectively improved HPV-related knowledge. Results support integrating chatbot technology into nursing education to enhance engagement and learning in sexual health topics.

Keywords Human papillomavirus, HPV, College students, Chatbot, Technology acceptance model, TAM

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Introduction

Human papillomavirus (HPV) infection remains one of the most prevalent sexually transmitted infections (STIs) globally, affecting individuals of all genders and posing a significant public health challenge [1, 2]. The incidence is particularly high among individuals in their late teens and early twenties, making college students a historically high-risk group [3]. Although the majority of HPV infections are asymptomatic and resolve spontaneously, persistent infections can lead to HPV-related conditions, including genital warts and various cancers such as cervical, vaginal, vulvar, penile, anal, oropharyngeal, and head and neck cancers [4, 5]. These outcomes underscore the critical importance of early prevention and comprehensive health education to reduce the disease burden. In addition to promoting HPV vaccination and safe sexual practices, cervical cancer screening remains a critical component of women's health. According to the American College of Obstetricians and Gynecologists [6], women aged 21 to 29 are recommended to undergo Pap smear screening every three years, while those aged 30 to 65 should receive co-testing with both Pap smear and HPV DNA testing to more comprehensively assess cervical cancer risk. In Taiwan, women aged 25 to 29 are advised to undergo Pap smear screening every three years. For women aged 30 and above, the national health program provides one free Pap smear per year, and free HPV DNA testing is offered at ages 35, 45, and 65 to improve the early detection of high-risk HPV types. By contrast, there are currently no routine HPV screening recommendations for men in the general population. However, HPV DNA testing may be considered for high-risk groups, such as men who have sex with men (MSM) or individuals presenting with clinical symptoms [7]. Common sampling sites include the genital and anal regions. According to data from the Centers for Disease Control and Prevention [8], gay and bisexual men have a 17-fold higher risk of developing anal cancer compared to heterosexual men, and the incidence of oropharyngeal cancer in men is approximately five times higher than in women. These findings underscore the importance of targeted HPV DNA testing for at-risk male populations to facilitate early diagnosis and timely medical intervention. However, numerous studies have shown that even nursing students, who are expected to have greater exposure to HPV-related knowledge than the general population, still demonstrate insufficient understanding of HPV infection and its prevention [9, 10].

Similarly, research in Taiwan has revealed comparable gaps in knowledge and awareness among nursing students [10]. This is particularly concerning, as these students will become frontline healthcare professionals who will serve in hospitals, community health centers, and schools, where they are expected not only to possess accurate and comprehensive health information but also to promote disease prevention and health education among the public. To fulfill these professional responsibilities effectively, nursing students must develop a more in-depth understanding of HPV infection and its prevention strategies. Strengthening HPV-related content in nursing education is thus crucial to improving students' knowledge and competence in this area [11]. Doing so not only enables them to safeguard their own health but also equips them to effectively disseminate accurate HPV prevention information to their future patients and communities.

With the rapid advancement of internet communication technologies and the widespread use of mobile devices, learners now have more opportunities to engage in interactive learning via smartphones and messaging applications [12]. Due to the limited coverage of HPV in formal curricula, students often have low baseline knowledge. Most rely on independent learning through the internet, self-directed searches, or short-term HPV prevention education activities [13, 14]. Although abundant information on HPV is available online, students may struggle to evaluate its credibility and accuracy without expert guidance [15]. In Taiwan, while HPV-related topics are briefly addressed in nursing education, the content is often limited in depth and is not presented as a dedicated module or chapter. Consequently, nursing students may acquire only superficial knowledge, lacking a comprehensive understanding of HPV infection, modes of transmission, preventive strategies, and vaccination. This curricular gap highlights the necessity of supplementary, self-directed learning tools to strengthen nursing students' knowledge and preparedness in HPV-related health promotion and care. In this context, chatbots have emerged as a promising digital tool to support self-directed health education by offering real-time, user-friendly, and flexible access to information [16]. For nursing students, chatbot-assisted learning may serve as a valuable supplement to classroom instruction. Chatbots are also regarded as an innovative modality in nursing education [17]. Although, despite the growing interest in chatbot-based learning, the extent to which students are willing to engage with such technology particularly in the absence of direct teacher facilitation remains uncertain [18]. Key concerns involve students' perceptions of the chatbot's usefulness, ease of use, and their overall attitude toward its integration into the learning process [19].

In the context of HPV prevention education, instructional delivery for nursing students has traditionally relied on lecture-based approaches, with limited exploration of innovative or technology-enhanced pedagogies. Nursing students are typically digital natives and are generally open to adopting new technologies, especially those that are user-friendly and offer immediate,

accurate, and private feedback [20, 21]. Chatbots are one such technology that aligns well with students' expectations and learning needs, particularly when addressing sensitive topics like HPV prevention. Given that TAM emphasizes perceived ease of use and perceived usefulness as key determinants of technology acceptance, it serves as an appropriate theoretical framework to explore students' engagement with a chatbot for sexual health education [20]. To date, no empirical studies have examined the application of chatbot-assisted learning in this area. Therefore, this study aimed to explore these uncertainties by examining nursing students' intentions to use a chatbot for HPV infection prevention education, the factors influencing their intention, and their actual usage behavior, based on the Technology Acceptance Model (TAM).

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), developed by Davis (1989) [22], is a widely recognized framework for explaining users' acceptance of new technologies. Adapted from the Theory of Reasoned Action (TRA) [23], TAM posits that two primary factors perceived usefulness (PU) and perceived ease of use (PEOU) influence users' behavioral responses toward technology adoption [24]. PU is defined as the degree to which a person believes that using a particular system would enhance their performance, whereas PEOU refers to the extent to which an individual believes that using the system would be free of effort [23]. These two beliefs affect the user's attitude toward using (ATU) the technology, which in turn influences their behavioral intention to use (BI), ultimately leading to actual use (AU). Specifically, PU has a direct effect on BI, while PEOU influences BI both directly and indirectly through PU. ATU reflects an individual's overall affective evaluation of the technology, BI indicates the user's intention or willingness to use it, and AU represents the actual usage behavior [22, 25].

Moreover, external variables such as age, gender, prior experience, and educational background may influence users' perceptions of PU and PEOU, thereby indirectly shaping their acceptance behavior [26, 27]. In this study, the selection of external variables was informed by previous research [28, 29] and adapted to the context of HPV prevention. Specifically, the external variables included participants' age, gender, prior experience with using a chatbot for learning purposes, family history of HPV-related diseases, sexual experience, number of lifetime sexual partners, and HPV vaccination status. Such variables were hypothesized to influence nursing students' PU and PEOU of the chatbot intervention. In the context of chatbot-assisted learning, users' acceptance of the technology is primarily driven by their PEOU and PU. A higher level of PEOU enhances PU, which subsequently influences their ATU the chatbot. This ATU affects their BI, ultimately leading to AU of the chatbot. These relationships provide the theoretical foundation for examining students' acceptance of chatbot-assisted HPV prevention education in this study.

Methods

Design and study participants

This study employed a cross-sectional research design. The participants were nursing students from a university in central Taiwan. Eligibility criteria included: (a) being a nursing student aged 18 to 25 and (b) the ability to converse in Mandarin or Taiwanese. Individuals with no prior exposure to any HPV-related information were excluded from the study, as confirmed verbally during the informed consent process. Because HPV content is briefly introduced in the nursing curriculum, most students met the inclusion criteria. Utilizing G*Power 3.1.9.2 software [30], a priori power analysis was conducted with an alpha level of 0.05, a power of 0.80, and nine predictors. The analysis indicated that a minimum sample size of 114 participants was required to detect a medium effect size ($f^2 = 0.15$). In total, 210 participants were recruited for this study, exceeding the minimum sample size requirement.

Instruments

Chatbot menu structure and content design

A chatbot was developed on the LINE platform to deliver self-directed HPV prevention education specifically designed for nursing students. The interface was mobile-responsive, user-friendly, and optimized for smartphone-based access. The HPV prevention chatbot included three primary menu options, each designed to support nursing students' self-directed learning and engagement: (Fig. 1):

Thematic modules: HPV

This section contained five educational modules aligned with current HPV-related health guidelines. Each module presented evidence-based content using short texts and visual aids to facilitate comprehension. The five topics included:

- Basic Knowledge of HPV: Introduced fundamental concepts such as HPV types, modes of transmission, and prevalence.
- HPV-Related Risks: Addressed key risk factors, including early sexual activity, multiple sexual partners, and the potential long-term consequences of persistent infection.
- HPV Prevention and Management: Provided behavioral prevention strategies and emphasized the

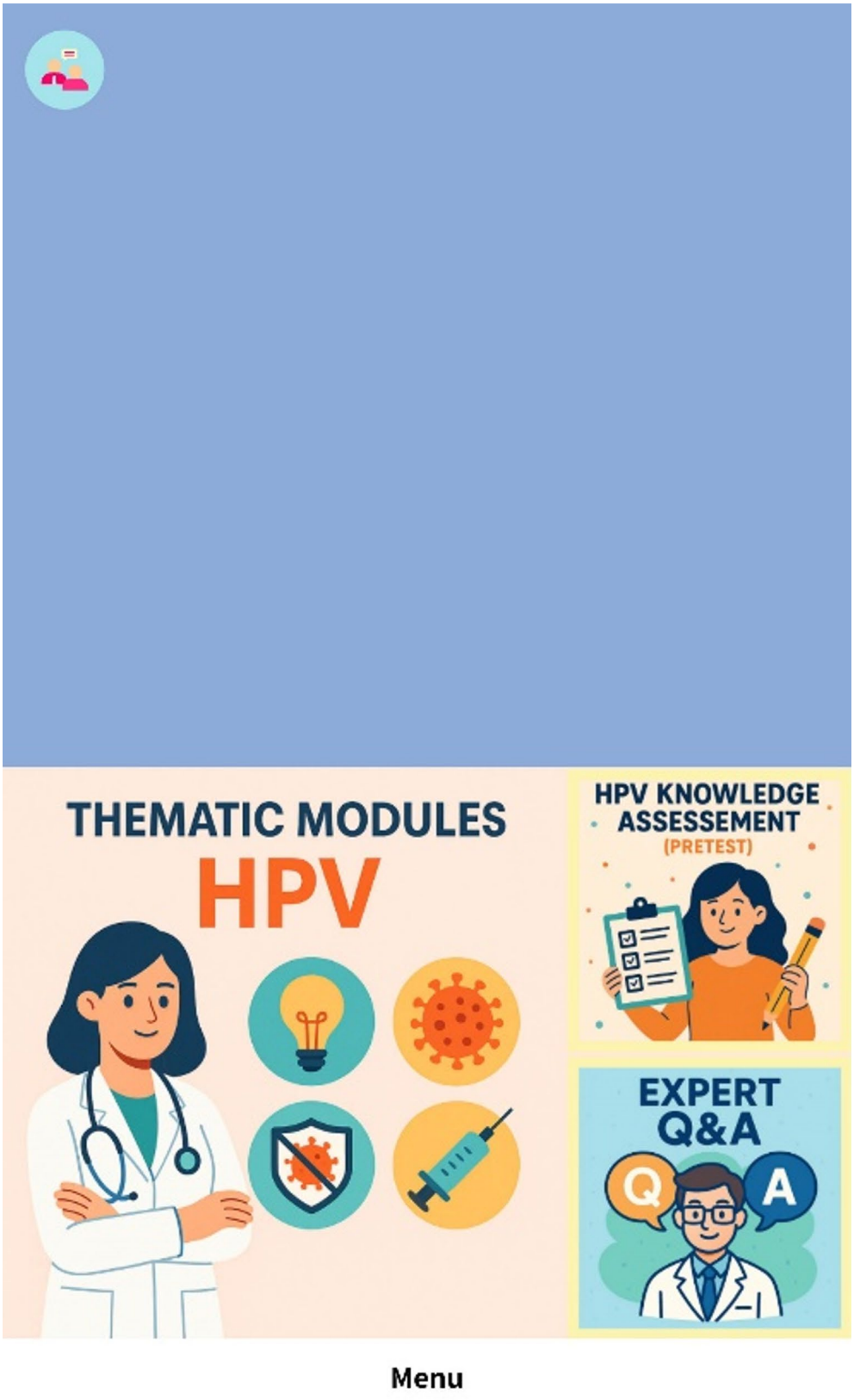

Fig. 1 Chatbot menu

importance of regular screening methods, such as Pap smears and HPV DNA testing.
- HPV Vaccination: Offered information on vaccine types, schedules, target populations, and clarified common misconceptions about safety and efficacy.
- Frequently Asked Questions (FAQs): Delivered automated responses to commonly asked questions through a keyword recognition system, employing conversational language to enhance accessibility and engagement.

To reinforce nursing students' HPV knowledge, the chatbot included an HPV Knowledge Assessment within its menu. Upon initial chatbot access, participants were guided by the researcher to complete a pretest consisting of 17 true/false questions, covering key concepts presented in the chatbot's thematic modules. Immediate feedback and correct answers were not provided at this stage to encourage participants' self-reflection and consolidation of knowledge. After two weeks, the chatbot automatically sent participants a link to complete the same assessment as a pretest, this time providing immediate feedback, correct answers, and concise explanations for each item. This pretest-posttest approach allowed for direct comparison of knowledge levels before and after the intervention. Details of the HPV knowledge assessment tool are described in the following section.

HPV knowledge assessment

This menu option allowed participants to access a 17-item true/false quiz at any time to self-assess their knowledge related to HPV infection, prevention, and vaccination. The instrument was developed based on previous HPV-related studies [3, 31, 32] and was aligned with the content of the thematic modules. The questionnaire covered essential topics such as HPV transmission, risk factors, prevention strategies, and vaccination. Upon completion, the system immediately provided participants with feedback, including correct answers and brief explanations, to facilitate learning and reinforce key concepts. Additionally, guidance on these topics was integrated into the Thematic Modules, allowing students to review relevant content and strengthen their understanding prior to and after completing the questionnaire.

Expert Q&A

This function allowed users to submit personalized questions directly through the chatbot interface. All inquiries were reviewed and answered by qualified healthcare professionals, ensuring evidence-based responses. This interactive component simulated real-time expert consultation. Real-time responses were available daily between 12:00 and 13:00. Inquiries submitted outside this time window were addressed on the following day.

Technology acceptance

The authors developed a technology acceptance scale for a nursing education chatbot based on relevant literature and the Technology Acceptance Model (TAM) [22, 28, 33]. The scale comprised five subscales: PU, PEOU, ATU, BI, and AU, with five items per subscale, totaling 25 items, with scores ranging from 25 to 100. Example items include:

PU: "Using the chatbot improves my learning efficiency."

PEOU: "The chatbot is easy to use."

ATU: "I have a positive attitude toward using the chatbot."

BI: "I intend to use the chatbot for learning in the future."

AU: "I used the chatbot frequently during the intervention."

All items were rated using a 4-point Likert scale, ranging from 1 (strongly disagree) to 4 (strongly agree), with higher scores indicating greater technology acceptance. Additionally, three open-ended questions were included to gather nursing students' feedback on chatbot usage. Content validity was confirmed by five experts in nursing education and technology, yielding a content validity index (CVI) of 0.88. Cronbach's α values indicated acceptable internal consistency across subscales: 0.85 (PU), 0.85 (PEOU), 0.78 (ATU), 0.82 (BI), and 0.82 (AU).

Demographic characteristics

The following variables were included: age, gender, year level, prior experience with using a chatbot for learning purposes, daily time spent using digital devices, family history of HPV-related diseases, sexual experience, number of lifetime sexual partners, and HPV vaccination status.

Ethical considerations and data collection

The study was approved by the institutional review board of the study setting (IRB No. 202500005A3). The data collection period was February to May 2025. With the support of faculty members at the study sites, the investigator introduced the study details to college students in classroom settings, clearly emphasizing their right to withdraw at any point without affecting their academic standing or learning privileges. Participants provided informed consent prior to involvement. Subsequently, nursing students received a QR code enabling access to the HPV prevention education chatbot. They also participated in a 20-minute training session that provided a comprehensive overview of the chatbot's functionalities and operational procedures. Participants engaged

with the chatbot using their smartphones for a two-week period. Following this period, participants completed an online questionnaire, taking approximately 10 to 15 min. To maintain data integrity and prevent duplication, IP tracking was employed, and participants were instructed to submit the questionnaire only once. All collected data were anonymized, coded, securely stored in password-protected digital files, and managed confidentially to ensure participants' privacy and anonymity.

Data analysis

Data were analyzed in May 2025. All statistical tests were performed using the SPSS 26.0 software (IBM Corp, Armonk, NY, USA). Data were summarized as means and standard deviations for continuous variables and as proportions for categorical variables. Independent samples t-tests and one-way ANOVA were conducted to examine differences in Technology Acceptance Model (TAM) constructs across demographic subgroups. Pearson correlation analysis was used to explore relationships among TAM variables. Linear regression analyses were performed to estimate TAM-related path coefficients. Additionally, open-ended responses and participant feedback were analyzed to supplement the quantitative findings.

Table 1 Descriptive statistics of participants' demographic characteristics and TAM constructs ($n = 210$)

Variable	*N* (%)	Mean ± SD
Age (year)		24.26 ± 7.06
Sex		
Male	20 (9.5)	
Female	190 (90.5)	
Year level		
First year	9 (4.3)	
Second year	46 (21.9)	
Third year	102 (48.6)	
Fourth year	53 (25.2)	
Experience of using a chatbot for learning purposes		
No Experience	128 (61.0)	
Have Experience	82 (39.0)	
Daily time spent using digital devices (including computers, smartphones, or pad)		
Less than 1 h	14 (6.7)	
1–3 h	51 (24.3)	
4–5 h	57 (27.1)	
More than 5 h	88 (41.9)	
Family history of HPV infection		
No	174 (82.9)	
Yes	36 (17.1)	
Sexual experience		
No	80 (38.1)	
Yes	130 (61.9)	
No. of lifetime sexual partners		
0	80 (38.1)	
1–2	40 (19.0)	
3–6	73 (34.8)	
≥ 7	13 (8.1)	
HPV vaccination status		
Non-vaccinated	115 (54.8)	
Vaccinated	95 (45.2)	

Results

Influence of external variables on TAM constructs among participants

The demographic characteristics of the nursing students are summarized in Table 1, serving as the basis for examining the influence of external variables on TAM constructs in subsequent analyses. A total of 210 students participated in the study, with a mean age of 24.26 years (SD = 7.06). The majority were female ($n = 190$, 90.5%), and only a small proportion were male ($n = 20$, 9.5%). Regarding academic year level, the largest group comprised third year students ($n = 102$, 48.6%), followed by fourth year ($n = 53$, 25.2%), second year ($n = 46$, 21.9%), and first year students ($n = 9$, 4.3%). A total of 82 participants (39.0%) reported having used chatbots for learning purposes, while 128 (61.0%) had no prior chatbot experience. In terms of digital engagement, most students reported using digital devices extensively: 88 students (41.9%) used them for more than 5 h per day, 57 (27.1%) for 4–5 h, 51 (24.3%) for 1–3 h, and 14 (6.7%) for less than 1 h daily.

A total of 130 students (61.9%) reported having had sexual experience, while 80 (38.1%) had not. When asked about the number of sexual partners, 80 (38.1%) reported none, 40 (19.0%) had 1–2 partners, 73 (34.8%) had 3–6 partners, and 13 (8.1%) had more than 7 partners. In addition, 36 (17.1%) students reported having a family history of HPV infection, while the majority ($n = 174$, 82.9%) did not. As for HPV vaccination status, 95 students (45.2%) reported having been vaccinated, whereas 115 (54.8%) had not received the HPV vaccine.

Correlations among five dimensions of using chatbot in HPV prevention education

Table 2 shows that the correlation coefficients among the five dimensions range from 0.86 to 0.94, indicating significant positive relationships. The range of scores for PU was 12 to 20, PEOU was 12 to 20, ATU was 14 to 20, BI was 13 to 20, and AU was 12 to 20. The mean scores were 16.70 (SD = 2.23) for PU, 16.73 (SD = 2.29) for PEOU, 16.85 (SD = 2.26) for ATU, 16.87 (SD = 2.27) for BI, and 16.78 (SD = 2.39) for AU (Table 2).

Acceptance of a chatbot in HPV prevention education

Table 3 presents the results of linear regression analyses exploring factors influencing nursing students'

Table 2 Correlations among five dimensions of using chatbot in HPV prevention education ($n = 210$)

Variables	Mean ± SD	Range	1	2	3	4	5
			r	*r*	*r*	*r*	*r*
1	16.70 ± 2.23	12–20	1				
2	16.73 ± 2.29	12–20	0.86**	1			
3	16.85 ± 2.26	14–20	0.87**	0.88**	1		
4	16.87 ± 2.27	13–20	0.90**	0.90**	0.94**	1	
5	16.78 ± 2.39	12–20	0.88**	0.87**	0.90**	0.93**	1

Row and column headings: 1 = Perceived usefulness, 2 = Perceived ease of use, 3 = Attitude toward using, 4 = Behavioral intention to use, 5 = Actual use

Pearson correlation coefficient. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

Table 3 Regression analysis of factors influencing chatbot acceptance in HPV prevention education ($n = 210$)

DV	R^2	IV	B	SE	β	95% CI	*t*	*p*
Perceived usefulness	0.746	Constant	17.379	0.553		16.288 to 18.469	31.430	< 0.001***
	($F = 99.187$, $p < 0.001$)	Age	-0.031	0.020	-0.097	-0.069 to 0.008	-1.544	0.124
		No Experience[a]	-1.510	0.288	-0.332	-2.077 to -0.943	-5.253	< 0.001*
		1–2 sexual partners[b]	1.127	0.390	0.199	0.357 to 1.896	2.886	0.004**
		3–6 sexual partners[b]	1.145	0.329	0.246	0.496 to 1.794	3.480	0.001**
		≥ 7 sexual partners[b]	0.537	0.537	0.066	-0.523 to 1.596	0.998	0.319
		Perceived ease of use	0.833	0.035	0.859	1.617 to 3.914	24.148	< 0.001***
Perceived ease of use	0.206	Constant	17.735	0.569		-16.613 to 18.856	31.175	< 0.001***
	($F = 10.56$, $p < 0.001$)	Age	-0.045	0.020	-0.138	-0.085 to -0.005	-2.209	0.028*
		No Experience[a]	-1.447	0.296	-0.308	-2.030 to -0.863	-4.890	< 0.001***
		1–2 sexual partners[b]	1.213	0.402	0.208	0.421 to 2.005	3.020	0.003**
		3–6 sexual partners[b]	1.167	0.339	0.243	0.500 to 1.835	3.448	0.001**
		≥ 7 sexual partners[b]	0.144	0.553	0.017	-0.946 to 1.235	0.261	0.794
Attitude toward using	0.822	Constant	1.173	0.513		0.161 to 2.185	2.286	0.023*
	($F = 477.22$, $p < 0.001$)	Perceived usefulness	0.423	0.058	0.417	0.309 to 0.538	7.291	< 0.001***
		Perceived ease of use	0.514	0.056	0.523	0.403 to 0.626	9.134	< 0.001***
Behavioral intention to use	0.908	Constant	0.134	0.375		-0.605 to 0.873	0.357	0.722
	($F = 1019.12$, $p < 0.001$)	Perceived usefulness	0.376	0.043	0.368	0.291 to 0.461	8.731	< 0.001***
		Attitude toward using	0.620	0.043	0.616	0.536 to 0.704	14.590	< 0.001***
Actual use	0.871	Constant	0.215	0.447		-0.666 to 1.095	0.480	0.632
	($F = 1398.35$, $p < 0.001$)	Behavioral intention to use	0.982	0.026	0.933	0.930 to 1.034	37.395	< 0.001***

Reference group: [a] Have Experience; [b] sexual partners

acceptance of chatbots for HPV prevention education. For PU, PEOU was the strongest positive predictor ($\beta = 0.859$, $p < 0.001$), explaining 74.6% of the variance. Students without prior chatbot learning experience reported significantly lower PU (β = -0.332, $p < 0.001$). Age was not a significant factor ($p = 0.124$). Regarding sexual partner history, students with 1–2 ($\beta = 0.238$, $p = 0.004$) and 3–6 partners ($\beta = 0.246$, $p = 0.001$) showed significantly higher PU, while those with ≥7 partners showed no significant effect ($\beta = 0.054$, $p = 0.319$). For PEOU, age (β = -0.138, $p = 0.028$) and chatbot experience (β = -0.308, $p < 0.001$) were significant predictors, accounting for 20.6% of the variance. In turn, ATU was significantly influenced by both PU ($\beta = 0.417$, $p < 0.001$) and PEOU ($\beta = 0.523$, $p < 0.001$), explaining 82.2% of the variance. BI was significantly influenced by PU ($\beta = 0.368$, $p < 0.001$) and attitude ($\beta = 0.616$, $p < 0.001$), accounting for 90.8% of the variance. Lastly, AU was strongly predicted by behavioral intention ($\beta = 0.933$, $p < 0.001$), with an explanatory power of 87.1%.

Feedback from open-ended questions

The feedback obtained from the 210 nursing student questionnaires was analyzed using conventional content analysis. Responses to the three open-ended questions were read repeatedly, coded, and grouped into categories. Positive comments indicated that the chatbot enabled students to "quickly find what I want to know," "learn faster than by reading textbooks," and "better understand complex topics." Minor limitations were also noted, such as "some answers felt too generic" and "needing a stable internet connection".

Discussion

The role of external variables in influencing chatbot acceptance

The demographic and behavioral profile of the participating nursing students provides essential context for interpreting their responses to the chatbot-assisted HPV prevention education. Age was significantly associated with PU and PEOU, with younger students reporting higher scores. This may reflect greater digital familiarity among younger users. Most participants were female and in their third year of study, consistent with typical nursing program demographics [17].

A substantial proportion of students (61%) had no prior experience using chatbots for learning, which may have influenced their initial perceptions of the technology. Prior studies have shown that familiarity with chatbots plays a critical role in shaping users' attitudes and acceptance. For example, Chang et al. (2022) found that the use of chatbots in nursing training significantly enhanced students' learning achievement and self-efficacy, suggesting that exposure to chatbot-based learning environments may promote deeper engagement [34]. Similarly, Hew et al. (2021) reported that students perceived chatbots as useful and easy to use in learning sessions, with prior interaction improving their behavioral engagement [35]. These findings support the notion that students with previous chatbot experience are more likely to perceive such tools as beneficial and user-friendly, likely due to greater comfort and familiarity with the interface and learning structure. These findings suggest that early exposure to chatbot-based tools can enhance acceptance and reduce barriers to engagement in nursing education. Participants reported varied levels of digital engagement, with over 40% using digital devices for more than five hours per day. This suggests that most students were digitally active and likely accustomed to engaging with technology in academic or personal contexts [36]. However, as indicated by TAM theory, general digital activity does not necessarily equate to higher PU or PEOU of a specific educational technology [22].

Sexual experience and the number of sexual partners were also documented, as these variables may reflect participants' personal relevance to HPV-related content. The relatively high percentage of students with sexual experience (61.9%) and with multiple sexual partners (42.9% reporting 3 or more) may suggest that HPV prevention is a pertinent issue for this population, potentially influencing motivation to engage with the educational intervention. Prior studies have highlighted that individuals who perceive themselves at risk for HPV are more likely to participate actively in prevention efforts [32, 37]. In addition to personal sexual history, family history of HPV infection may also contribute to participants' engagement with HPV prevention education. In this study, 17.1% of students reported having a family member with an HPV infection history. This subgroup may have a heightened awareness of HPV-related health issues and be more receptive to educational content addressing prevention strategies. Familial experiences with HPV may personalize the topic, increasing the perceived relevance of the chatbot intervention. Previous studies have indicated that individuals with a family history of HPV infection often exhibit a greater motivation to seek health-related information and engage in preventive behaviors [38]. For nursing students, such personal connections to HPV may translate into a more proactive attitude toward learning and using digital health tools.

Additionally, nearly half of the students (45.2%) reported having received HPV vaccination, indicating a moderate level of preventive health behavior within the group. However, the vaccination rate also reveals that a significant portion of students remain unvaccinated despite their future role as health educators. This gap underscores the importance of integrating HPV-related education into nursing curricula and leveraging innovative tools such as chatbots to address knowledge gaps and influence attitudes toward prevention.

Interrelationships among TAM constructs in the context of chatbot use

The findings of this study revealed strong positive correlations among the five dimensions of the Technology Acceptance Model (TAM), with correlation coefficients ranging from 0.86 to 0.94. These results suggest a high degree of internal consistency and conceptual alignment across PU, PEOU, ATU, BI, and AU. Such high intercorrelations are consistent with previous TAM-based research in digital health and education settings, indicating that students' evaluations of a technology's utility, ease of use, and personal attitudes are closely intertwined [39–41]. These findings further support the growing consensus that chatbots have the potential to be used in nursing education [17].

The mean scores across all constructs were relatively high ranging from 16.70 to 16.87 out of a maximum of 20 indicating that most nursing students held favorable perceptions toward the chatbot-assisted HPV prevention education. These findings support the notion that when users perceive a technology to be easy to use and useful, they are more likely to develop a positive attitude toward its use and form stronger behavioral intentions, which ultimately leads to actual engagement [22, 42].

Overall acceptance of the chatbot and its educational implications

The regression analyses provide strong empirical support for the Technology Acceptance Model (TAM) in the context of chatbot-assisted HPV prevention education.

PEOU emerged as the most significant predictor of PU, highlighting the importance of intuitive design in fostering positive evaluations. Prior experience with chatbots was also a key factor; students without previous exposure reported significantly lower levels of PEOU, suggesting that familiarity plays a critical role in shaping technology acceptance.

Sexual behavior variables demonstrated a nuanced influence on PU. Students who reported having 1–2 or 3–6 sexual partners showed significantly higher PU scores compared to those with no sexual experience, suggesting that moderate levels of sexual activity may increase the perceived relevance of HPV prevention content. Interestingly, participants with ≥7 sexual partners did not show a significant difference. One possible explanation is that individuals with a higher number of sexual partners may experience risk normalization or desensitization, reducing their perceived need for preventive education [43, 44].

The model further confirmed that PU and PEOU positively influenced ATU the chatbot, which in turn strongly predicted BI. Ultimately, BI was the only significant predictor of AU, accounting for 87.1% of its variance. These findings support the sequential relationships proposed by the TAM and highlight the feasibility of implementing chatbot interventions in nursing education to enhance student engagement in HPV prevention education.

Qualitative insights into user perceptions and suggestions

The open-ended feedback provided by nursing students offers valuable insight into their perceptions of using a chatbot for HPV prevention education. Overall, students expressed positive experiences, frequently noting that the chatbot enabled them to "quickly find what I want to know" and "learn faster than by reading textbooks." These responses highlight students' appreciation for the chatbot's immediacy, accessibility, and ability to simplify complex concepts qualities that are consistent with prior research on the benefits of conversational agents in education [45, 46]. Notably, students also reported that the chatbot facilitated their understanding of difficult topics, reinforcing the role of AI-based tools as effective supplements to traditional learning methods. This supports findings that chatbots can address frequently asked questions and guide learners through complex content via interactive dialogues [47, 48].

Despite the overall positive reception, students also identified areas for improvement. Some described the chatbot's responses as "too generic," indicating a need for more context-specific or personalized answers. Others mentioned that stable internet connectivity was necessary for optimal use, highlighting a technical limitation that may affect accessibility, particularly in resource-limited environments. These concerns underscore the importance of iterative development and user-centered design in chatbot implementation, ensuring content quality and technical reliability. The integration of student feedback into future versions of the chatbot could improve its relevance, responsiveness, and usability. Incorporating natural language processing improvements and adaptive content delivery may help address the identified limitations and further increase user satisfaction.

Implications for nursing practice

This study supports the feasibility of chatbot-assisted education in enhancing nursing students' understanding of HPV prevention. Integrating such technology into nursing curricula may promote engagement in sexual health education and better prepare students for digital healthcare environments [34, 35]. Although the current study focused on student education, chatbots may also hold potential for broader use in patient education, particularly in contexts requiring privacy, accessibility, and consistent information delivery [47, 48]. The findings of this study have important implications for nursing education and curriculum design. Chatbot-based interventions may serve as flexible and scalable tools to complement traditional lecture-based instruction. They can support self-paced learning, reinforce key concepts, and provide timely, personalized feedback particularly in areas involving sensitive health topics such as HPV prevention. Educators can integrate chatbots into pre-class preparation, interactive assignments, and debriefings to promote engagement and autonomy in learning.

This study contributes to the growing literature on chatbot use in health professions education. Compared to traditional digital tools, chatbots provide real-time interaction, immediate feedback, and personalized responses, making them ideal for addressing sensitive topics such as HPV prevention. They also support self-directed learning while ensuring privacy. In addition, chatbots are low-cost, accessible, and reusable, which makes them practical for diverse educational settings. Recent studies have shown that chatbot-assisted learning can improve student engagement, knowledge acquisition, and self-efficacy [17, 18, 34]. By applying the Technology Acceptance Model, this study offers a framework for understanding students' acceptance of chatbot technology and its integration into nursing education.

Strength and limitations

This study has several strengths. First, it is one of the few to apply the Technology Acceptance Model (TAM) to examine nursing students' acceptance of a chatbot for HPV prevention education. Second, the TAM-based scale showed good internal consistency, ensuring reliable measurement. Third, the inclusion of chatbot usage experience and HPV-related behaviors (such as family

history of HPV-related diseases, sexual experience, number of lifetime sexual partners, and HPV vaccination status) as external variables provided important context for understanding students' acceptance of the technology. Nonetheless, the study has limitations. Its cross-sectional design limits the ability to draw causal conclusions. In addition, the sample was recruited from a single region in Taiwan and consisted mostly of female students, which may affect the generalizability of the results. Future studies should consider longitudinal or experimental designs, recruit more diverse samples, and include objective usage data to strengthen model validation and support the development of chatbot-based health education.

This study was conducted in a specific cultural and linguistic context among nursing students in Taiwan and the intervention was delivered in Mandarin. These contextual factors, including cultural norms, language preferences, and variations in nursing education, may influence students' perceptions and acceptance of chatbot-assisted learning. Therefore, the transferability of the findings to other populations and settings may be constrained. Future studies are encouraged to adopt longitudinal or experimental designs, include more diverse and international samples, and incorporate objective usage data to enhance model validation and support the broader application of chatbot-based health education tools.

Conclusion

This study demonstrated the applicability of the Technology Acceptance Model in exploring nursing students' acceptance of a chatbot designed for HPV prevention education. The findings revealed that PU and PEOU significantly influenced students' attitudes and behavioral intentions, ultimately affecting their AU of the chatbot. Qualitative feedback further supported the quantitative results, highlighting the chatbot's effectiveness in enhancing understanding and learning efficiency. While most students responded positively, minor limitations such as generic responses and reliance on internet connectivity were noted. These results underscore the potential of chatbot-assisted education in nursing curricula and suggest that continued refinement of content and technical infrastructure could further improve user engagement and learning outcomes.

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Author contributions
C.-Y.L. and W.-S.H. contributed meaningfully to this study. All authors have read and agreed to the published version of the manuscript.

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Data availability
Data available on request due to restrictions regarding privacy, legal and ethical concerns. The data presented in this study are available on request from the corresponding author.

Declarations

Ethical approval
The procedures employed in this study complied with the ethical standards of the relevant institutional and national research committees and adhered to the principles of the 1964 Declaration of Helsinki and its subsequent amendments or equivalent ethical guidelines. The study was approved by the Institutional Review Board (IRB) of Jen-Ai Hospital (JAH) under reference number 202500005A3, with approval granted on February 25, 2025. All participants were fully informed about the study's objectives, procedures, and their rights, and written informed consent was obtained prior to participation.

Informed consent
Every participant in the study provided their explicit informed consent.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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